PROJECT U1–U2 TIME SERIES

Canada's Health Profile

Exploratory Data Analysis of WHO Indicators

Population

Life expectancy

Health risks

Time series

SUBJECT
Time Series

STUDENT
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October 12, 2026



Data source: World Health Organization (WHO)

MCC · TecNM Cd. Guzmán

A STUDY OF WORLD HEALTH ORGANIZATION DATA

Canada's Health Profile

How the health of Canadians has changed, 1950–2024

9 indicators · 4 topics

POPULATION

13.7M → 39.3M

The population keeps growing, but **more slowly**: from about 3% a year to about 1% a year.

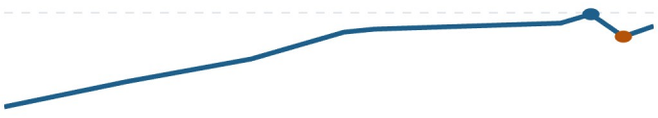


Fastest growth: 1957 (baby boom) **Slowest growth: 2021**

LIFE EXPECTANCY

79.1 → 82.0 years

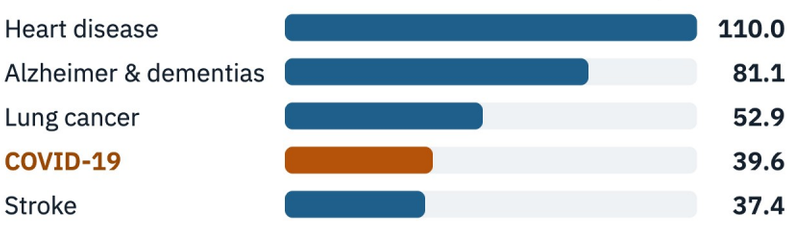
Canadians are living longer, but gains have **slowed down near 82 years**. Women still outlive men, though the gap is shrinking.



2012–2018: almost no change **2020: drop due to COVID-19**

WHAT CANADIANS DIE OF · 2021

deaths per 100,000



7 of the top 10 causes are long-term illnesses, which is common in a population that lives longer and is **getting older**.

HOW THESE INDICATORS CHANGE



Each year vs. the year before

Slow and steady: they change over decades, not overnight.

Predictable: each year looks a lot like the year before.

No repeating ups and downs from season to season.

Few surprises: only 1971 and 2020–21 stand out.

HEALTH RISKS: WHAT IMPROVED AND WHAT GOT WORSE

High blood pressure

33.0% → 22.1%

Adults 30–79 · 1990–2019

↓ Improved

Tobacco use

28.9% → 11.2%

Age 15+ · 2000–2024 · expected 8.9% by 2030

↓ Improved

Obesity

13.96% → 26.2%

Adults 18+ · 1990–2022 · nearly doubled

↑ Worsened

Alcohol

9.4 → 10.3 → 10 L

Litres per person · highest in 2008

→ Stable

Safe drinking water

99.05% → 96.9%

Share of people · 2000–2024 · falling since 2004

↘ Slowly declining

One unexpected event: COVID-19

0.74% slowest population growth since 1951 (2021)

−0.72 yrs drop in life expectancy in 2020, with signs of recovery in 2021

#4 most common cause of death in 2021

WHAT WE LEARNED

1. **Change is slowing down.** Population growth, life expectancy gains and most health risks are changing more slowly.
2. **An older population.** More people will need care for long-term illnesses.
3. **Prevention helps, but not everywhere.** Smoking and high blood pressure went down; obesity nearly doubled.
4. **Unexpected events matter.** COVID-19 caused a sharp but temporary drop in population growth and life expectancy.

Source: WHO, Health data overview for Canada (data.who.int/countries/124). Analysis in Python

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Series de Tiempo · MCC · TecNM Cd. Guzmán

Canada's Health Profile: Exploratory Data Analysis of WHO Indicators

1. Executive Summary

This report explores how the health of Canada's population has changed over the last decades. It uses public data from the World Health Organization (WHO) and looks at nine annual indicators grouped in four blocks: **population**, **life expectancy**, **leading causes of death**, and **risk factors and environment** (hypertension, alcohol, obesity, safe drinking water and tobacco). The goal is not to forecast, but to describe each series: its trend, whether it has seasonality or cycles, its unusual years, and how its behaviour changes over time.

Data and source

All files were downloaded from the WHO country page "Health data overview for Canada" (https://data.who.int/countries/124). Each indicator comes as a CSV file that contains every country; The population file is an Excel table with the population by single age (0 to 100+), in thousands. All series are **annual**.

Indicator	Unit	Period	Observations
Total population	Millions (sum of all ages)	1950-2023	74 years
Annual population growth rate	% change vs. previous year	1951-2023	73 years (derived)
Life expectancy at birth	Years (female, male, both)	2000-2021	22 years x 3 groups
Causes of death	Deaths per 100,000 population	2021	134 causes (one year)
Hypertension, adults 30-79	% of adults	1990-2019	30 years
Alcohol consumption, age 15+	Litres of pure alcohol per person	2000-2022	23 years
Obesity, adults 18+	% of adults	1990-2022	33 years
Safely managed drinking water	% of population	2000-2024	25 years
Tobacco use, age 15+	% of people	2000-2024	9 years (irregular) and WHO projections to 2030

Table 1. Datasets used in the analysis.

Methods

The analysis was carried out in Python. The data were loaded and prepared with pandas: each file was filtered to keep the records for Canada.

Each indicator was then explored with the same set of techniques. Time plots display each series against time and were used to identify trend, changes in direction and unusual years. Summary statistics (minimum, percentiles, median, mean and maximum), computed with tsfeatures, describe the level and spread of a series. Lag plots compare each observation with the value observed k years earlier, for lags 1 to 9, and were used to assess time dependence and seasonality. In the case of *Leading causes of death*, since data were available for only a single year, they were summarized with a ranked bar chart. All figures were produced with plotly and utilsforecast.

Why the analysis is relevant

These indicators describe how many people the health system has to serve, how long they live, what they die of, and which risks are growing or shrinking. Read together, they help a decision-maker see where prevention has worked, where it has not, and how much a single unexpected event can change the picture.

Main findings

- **Population** keeps growing, but more slowly. It rose from 13.7 million (1950) to 39.3 million (2023), while the annual growth rate fell from almost 3 % to about 1 %.
- **Life expectancy increased and is stabilizing.** It went from 79.1 years (2000) to 82.0 (2019), with a single drop of 0.72 years in 2020.
- **Chronic diseases lead mortality**. In 2021 the top causes were ischaemic heart disease (110.0 deaths per 100,000), dementias (81.1) and lung cancers (52.9).
- **Hypertension and tobacco use improved**. Hypertension fell from 33.0 % to 22.1 % (1990-2019) and tobacco use from 28.9 % to 11.2 % (2000-2024).
- **Obesity worsened**. Adult obesity almost doubled, from 13.96 % (1990) to 26.2 % (2022), although the increase has slowed.
- **Alcohol consumption stabilized**. It peaked at 10.3 litres per person in 2008 and has stayed close to 10 litres since 2014.
- **Safe drinking water is high but declining slowly**. Coverage went from 99.05 % (2000) to 96.9 % (2024).
- **No series shows seasonality**. All are dominated by trend; unusual years appear only in the growth rate (1971, 2021) and in life expectancy (2020).

Limitations

With annual data, patterns inside a year cannot be observed. According to the WHO metadata, several indicators are official estimates whose values are adjusted or predicted by a statistical model, not direct yearly measurements.The causes of death are available for one year only, so they are described as a ranking and not as a time series. Tobacco has few, unevenly spaced observations, so lag plots were not applied to it.

2. Exploratory Data Analysis

2.1 Population

Two series were built from the population file: the **total population** (sum of all ages) and its **annual growth rate** (percentage change against the previous year). The analysis focuses on the population growth rate, since total population increases steadily over the period and therefore masks changes in the pace of growth.

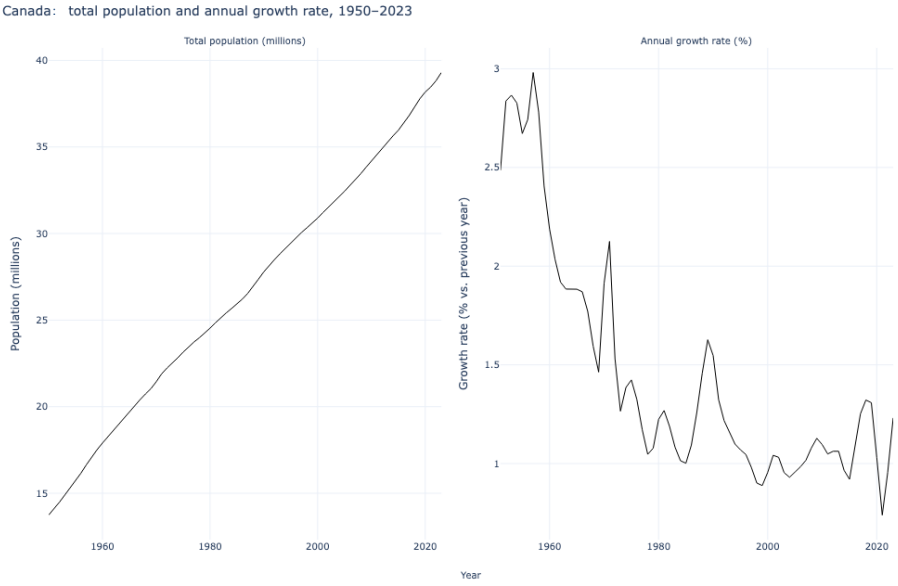


Figure 1. Total population (left) and annual growth rate (right), Canada 1950–2023.

The total population has a clear **positive trend**: it almost tripled, from 13.7 to 39.3 million. However, the growth rate has a **negative trend**: it was between 2.5 % and 3 % in the 1950s and has stayed close to 1 % since the 1990s. This means that the population keeps growing, but each year a little more slowly.

Series	Min	P25	Median	Mean	P75	Max
Annual growth rate (%)	0.74	1.05	1.22	1.45	1.77	2.98

Table 2. Summary statistics of the annual growth rate, 1951–2023.

Half of the years had a growth rate between 1.05 % and 1.77 %. The maximum is 2.98 % (1957) and the minimum is 0.74 % (2021).

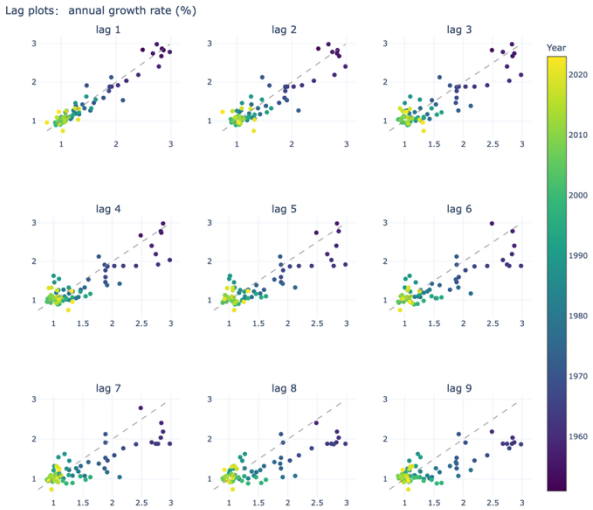


Figure 2. Lag plots of the annual growth rate.

Lag plots. At lag 1 the points are close to the diagonal: the growth rate of one year is similar to that of the previous year (positive autocorrelation). From lag 2 onwards the cloud opens, so the similarity fades as the gap widens. At lags 6 to 9 most points fall below the diagonal, which confirms the negative trend. The 1950s cluster at the top (2.4-3%), reflecting the post-war baby boom, while years from 2000 onwards cluster near 1%, where the rate has stabilized. The main outliers are 1970-1971 (a jump from 1.46% to 2.13%) and 2021 (0.74%, the series minimum, likely due to COVID-19). No lag stands out from the others, so there is no seasonality.

2.2 Life expectancy

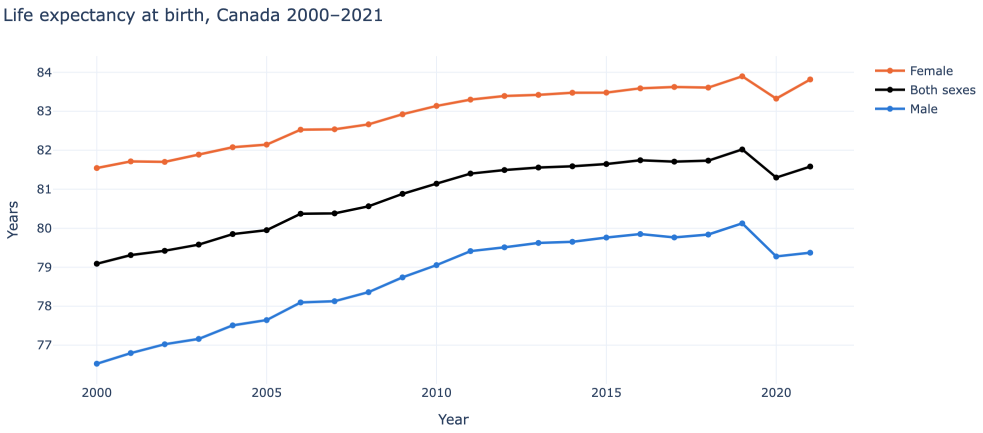


Figure 3. Life expectancy at birth by sex, Canada 2000–2021.

The three lines have a **positive trend** from 2000 to 2019: life expectancy for both sexes went from 79.1 to 82.0 years, about half a year gained every four years. The pattern changes around 2011: between 2012 and 2018 the value barely moved (81.5 to 81.7 years), so the improvement slowed down as it got close to 82. Women always live longer than men, but the gap narrowed from 5.0 years in 2000 to 3.8 in 2019.The only **outlier** is **2020**: life expectancy fell 0.72 years (from 82.0 to 81.3), the only large drop in the series, and it coincides with COVID-19. One year took the indicator back to about its 2011 level. The fall was larger for men (0.84 years) than for women (0.58), and 2021 shows a partial recovery (81.6).

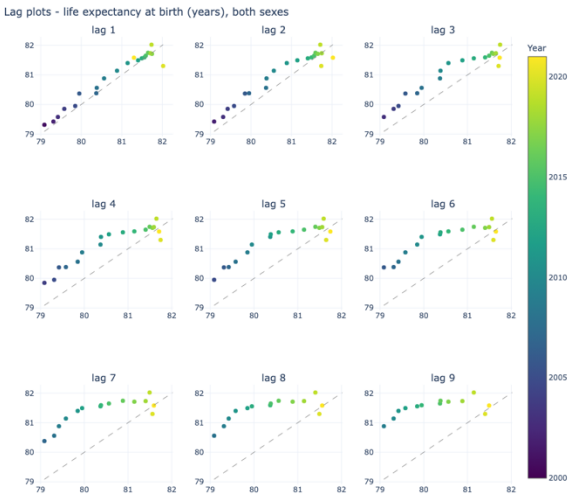


Figure 4. Lag plots life expectancy at birth (years) both sexes.

Lag plots. At lags 1 to 3 (and partly lag 4) the points are close to the diagonal: life expectancy in one year is very similar to that of the previous years (positive autocorrelation). As the lag increases, the similarity weakens. At lags 6 to 8 the cloud lies clearly above the diagonal, which confirms the positive trend, driven by advances in medicine, health research and public policy. Between 2012 and 2018 the points form an almost flat band (81.5 to 81.7 years), showing that the increase slows as life expectancy approaches 82 years. The main outliers are 2020 and 2021, which fall below the diagonal at every lag. No lag stands out from the others, so there is no seasonality or cycles.

2.3 Leading causes of death

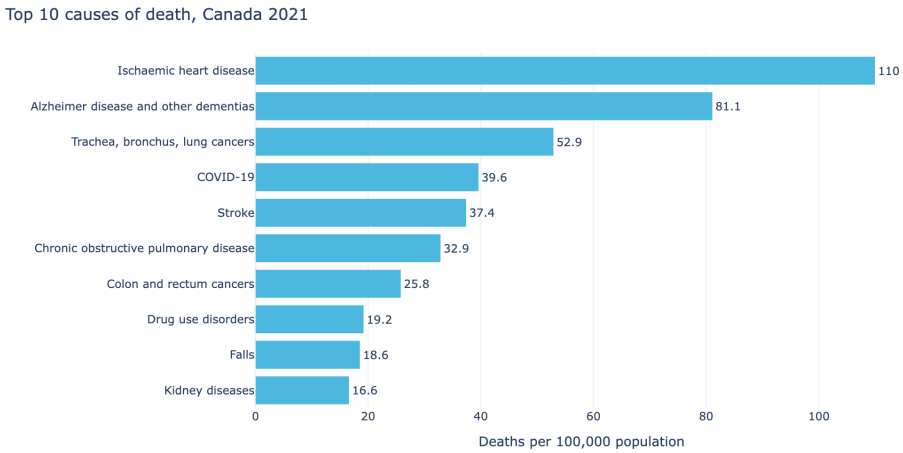


Figure 5. Top 10 causes of death, Canada 2021 (deaths per 100,000 population).

Leading causes of death. Since data are available for a single year (2021) across 134 causes, this indicator is analysed as a ranking. Ischaemic heart disease is by far the leading cause of death (110.0 deaths per 100,000), followed by Alzheimer disease and other dementias (81.1) and trachea, bronchus and lung cancers (52.9). COVID-19 ranks fourth (39.6), just ahead of stroke (37.4). Seven of the ten leading causes are chronic conditions (heart disease, dementias, cancers, stroke, and lung and kidney diseases), a pattern consistent with a long-lived, ageing population. The remaining three are of a different nature: COVID-19, drug use disorders (19.2) and falls (18.6).

2.4 Risk factors and environment

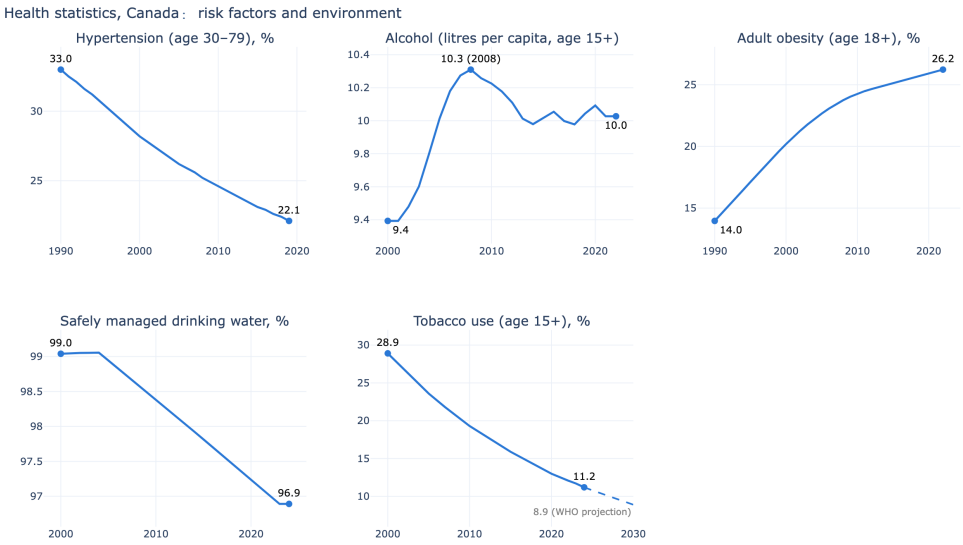


Figure 6. Health indicators for Canada.

Indicator	Period	Change	Trend	Seasonality / outliers
Hypertension (30–79)	1990–2019	33.0% to 22.1%	Negative, steady	No / No
Alcohol (L per person 15+)	2000–2022	9.4 to 10.3 (2008) to 10.0	Up, then stable	No / No
Adult obesity (18+)	1990–2022	13.96% to 26.2%	Positive, slowing	No / No
Safe drinking water	2000–2024	99.05% to 96.9%	Slightly negative	No / No
Tobacco use (15+)	2000–2024	28.9% to 11.2%	Negative, strong	Not testable / No

Table 3. Summary of the five indicators.

- **Hypertension.** Prevalence among adults aged 30-79 fell from 33.0% (1990) to 22.1% (2019), a clear negative trend, declining faster in the 1990s than in the 2010s.
- **Alcohol consumption.** Consumption rose from 9.4 L (2000) to a peak of 10.3 L (2008), then declined and has remained stable at around 10 L since 2014, so the trend changed direction over the period.
- **Obesity.** Prevalence nearly doubled, from 13.96% (1990) to 26.2% (2022). The increase has slowed since the 1990s but never reversed, showing a persistent positive trend.
- **Safely managed drinking water.** Coverage remained at about 99.05% from 2000 to 2004, then declined steadily to 96.9% in 2023-2024.
- **Tobacco use.** Use among people aged 15+ fell from 28.9% (2000) to 11.2% (2024), more slowly in recent years; the WHO projects 8.9% by 2030. Lag plots were not produced because observations are irregularly spaced, but the series shows a strong, consistent negative trend with no unusual years.

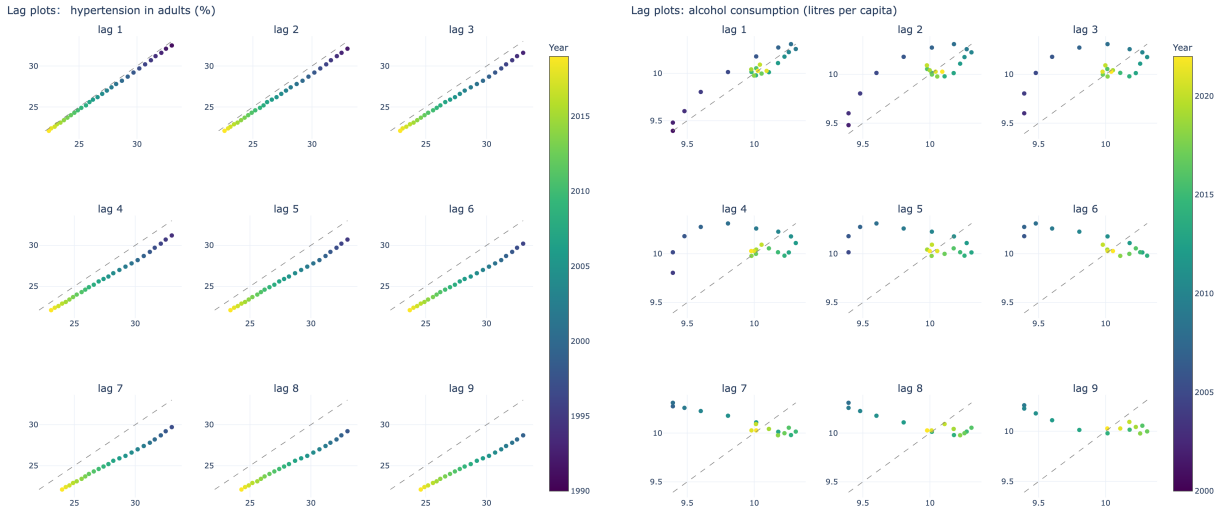


Figure 7. Lag plots of hypertension (left) and alcohol consumption (right).

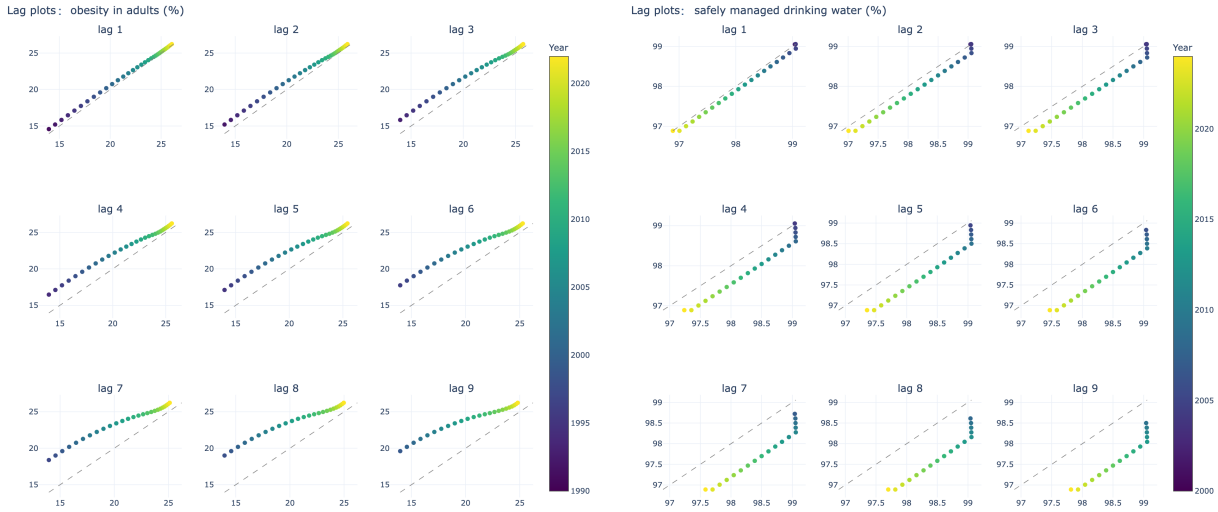


Figure 8. Lag plots of adult obesity (left) and safely managed drinking water (right).

Hypertension. At every lag the points lie below the diagonal, showing strong positive autocorrelation and a steady negative trend (33.0% to 22.1%) that slows in recent years. No seasonality or outliers. **Alcohol consumption**. Points follow the diagonal at lags 1 to 3, but flatten from lag 4 and slope downward from lag 5, reflecting the rise to 10.3 L in 2008 and later stabilization around 10 L. No cycle or seasonality. **Obesity.** At every lag the points lie above the diagonal, showing strong positive autocorrelation and a persistent positive trend (13.96% to 26.2%) that is slowing but has not reversed. No seasonality or outliers. **Safely managed drinking water**. Strong positive autocorrelation at every lag, with points falling further below the diagonal as the lag increases, confirming a gradual decline after 2004 (to about 97%). No seasonality or outliers.

3. Conclusions

The exploratory analysis shows that Canada's health indicators **change slowly and steadily** over decades, with few year-to-year fluctuations. All the annual series examined are dominated by trend, and **none shows seasonality or cycles**. The lag plots confirmed **strong positive autocorrelation** at short lags in every indicator: the value of one year is a good guide to the next. For series that changed direction or slowed down, such as the population growth rate, life expectancy and alcohol consumption, this similarity fades at longer lags, so the distant past becomes a poor predictor of the present.

A common pattern across indicators is **deceleration**. The population keeps growing, but its annual growth rate fell from almost 3% in the 1950s, during the post-war baby boom, to about 1% since the 1990s. Life expectancy rose from 79.1 to 82.0 years between 2000 and 2019 but has levelled off near 82. The decline in hypertension and tobacco use, and the increase in obesity, have also slowed in recent years.

Taken together, slower growth and longer lives describe an **ageing population**, and the mortality ranking for 2021 is consistent with this picture: seven of the ten leading causes of death are chronic conditions, led by ischaemic heart disease and dementias, and falls also appear in the top ten. For the health system, this implies a growing demand for the management of chronic disease, long-term care and services for older adults.

The risk factors show where prevention has and has not worked. **Tobacco use** (28.9% to 11.2%) and **hypertension** (33.0% to 22.1%) improved steadily, which suggests that sustained public health policies, detection and treatment have been effective. In contrast, **obesity** nearly doubled and is the only risk factor that clearly worsened. **Alcohol consumption** increased until 2008 and then stabilized around 10 litres per person, without returning to its 2000 level. **Safe drinking water** coverage remains high, but its gradual decline since the mid-2000s deserves attention, especially in rural and remote communities.

Finally, the analysis shows how much a single **unexpected event** can alter the picture. **COVID-19** appears in three separate indicators: it caused the lowest population growth rate in the series (0.74% in 2021), a 0.72-year drop in life expectancy in 2020 that returned the indicator to roughly its 2011 level, and it ranked as the fourth leading cause of death in 2021. Apart from 1970-1971 in the growth rate, these are the only unusual years in the data.

In summary, the dataset shows that Canada has made clear **progress in life expectancy** and in several preventable risks, but faces the challenges of an ageing population, rising obesity and the vulnerability of its health indicators to unexpected events.

4. Appendices

The original dataset and the complete Python script used for the analysis are available in the following repository: https://github.com/Ana-Sofi/canada-eda